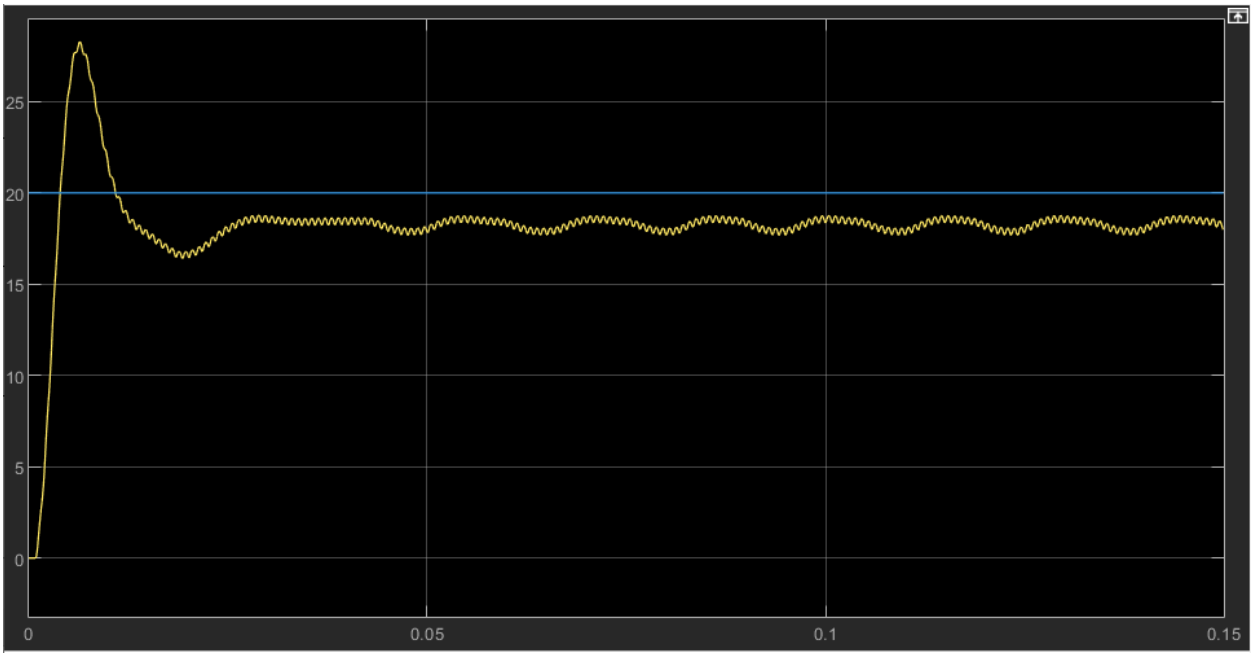
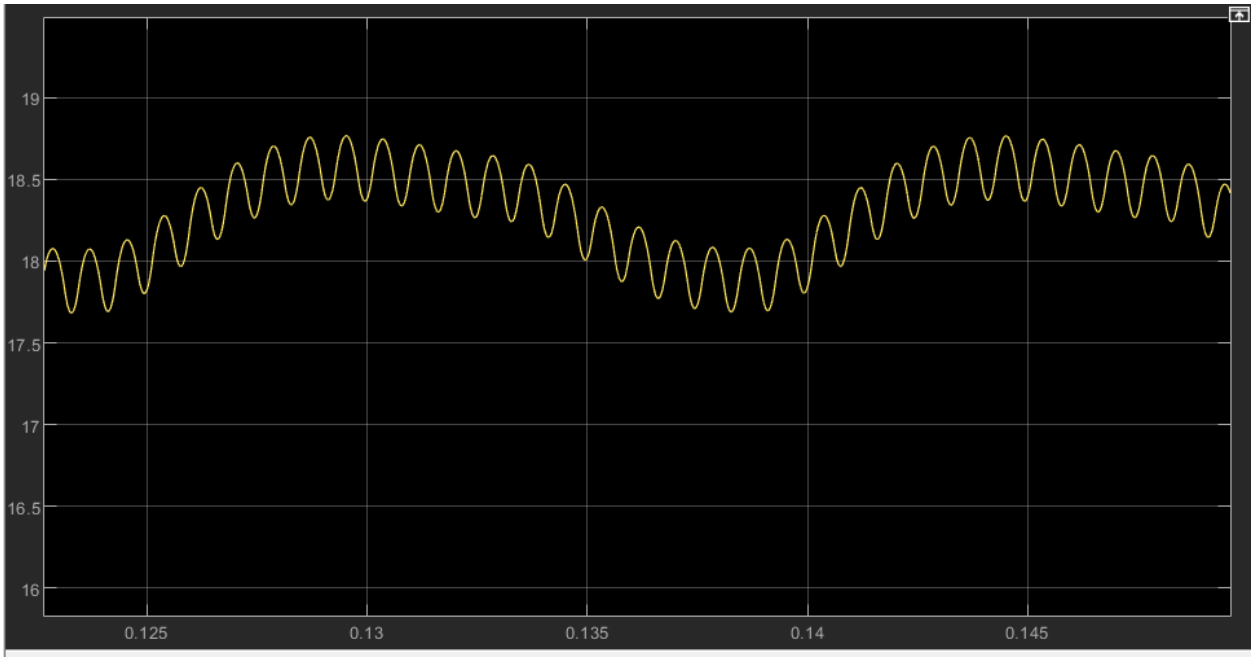
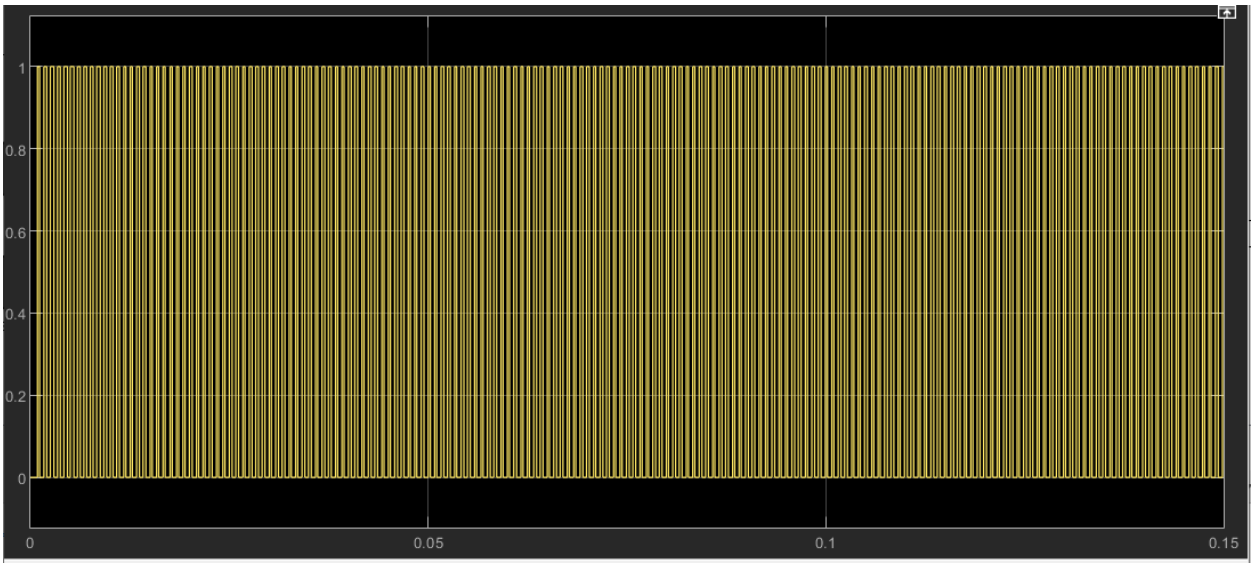
Mastan Andrei-Mircea

PE Project-Part 5

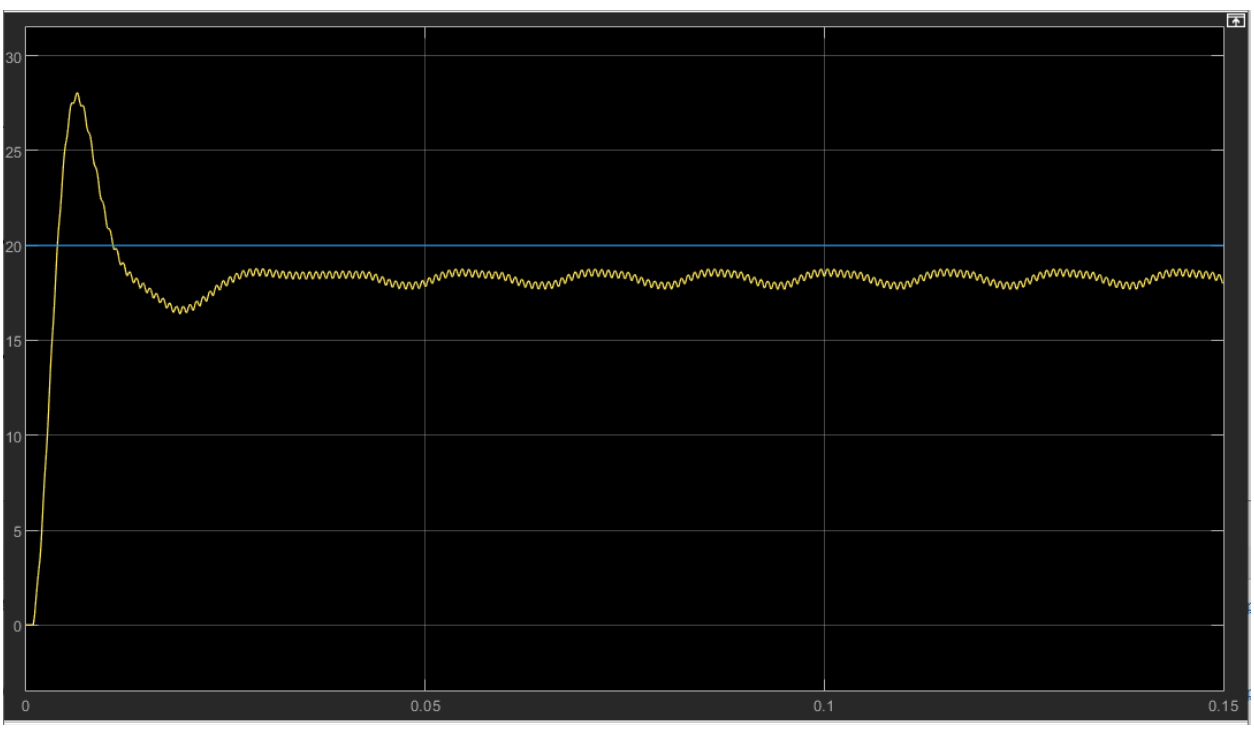


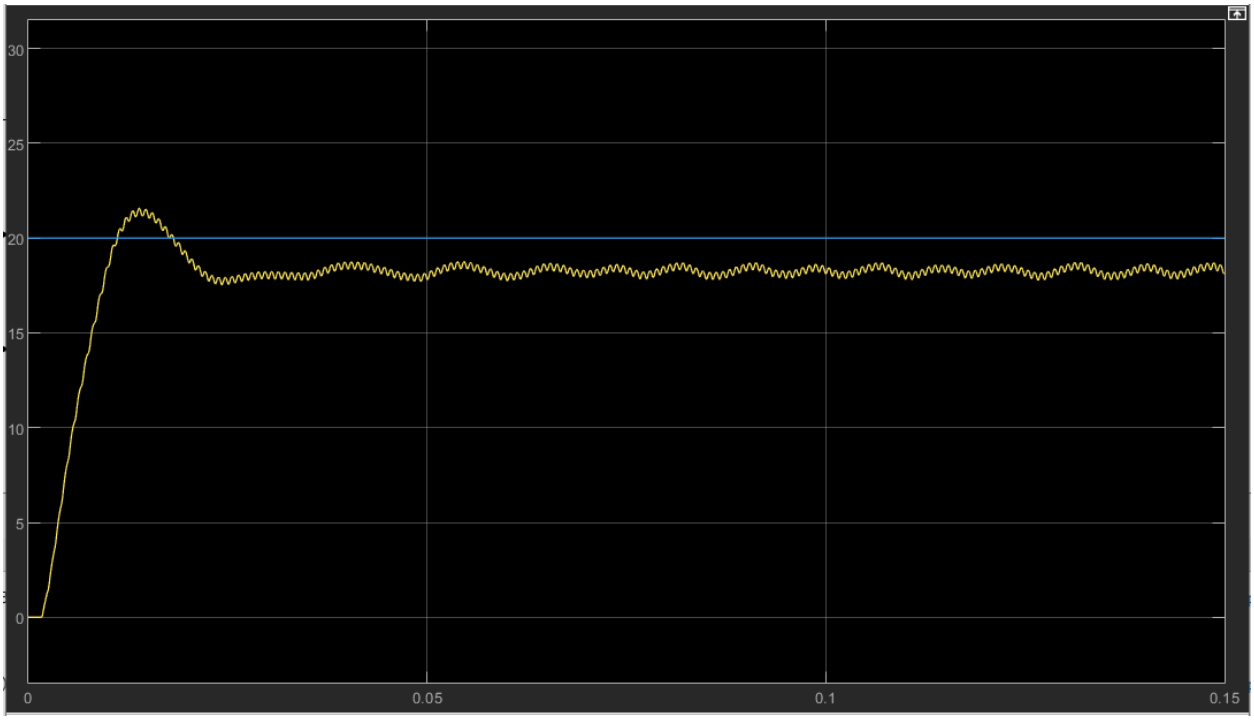


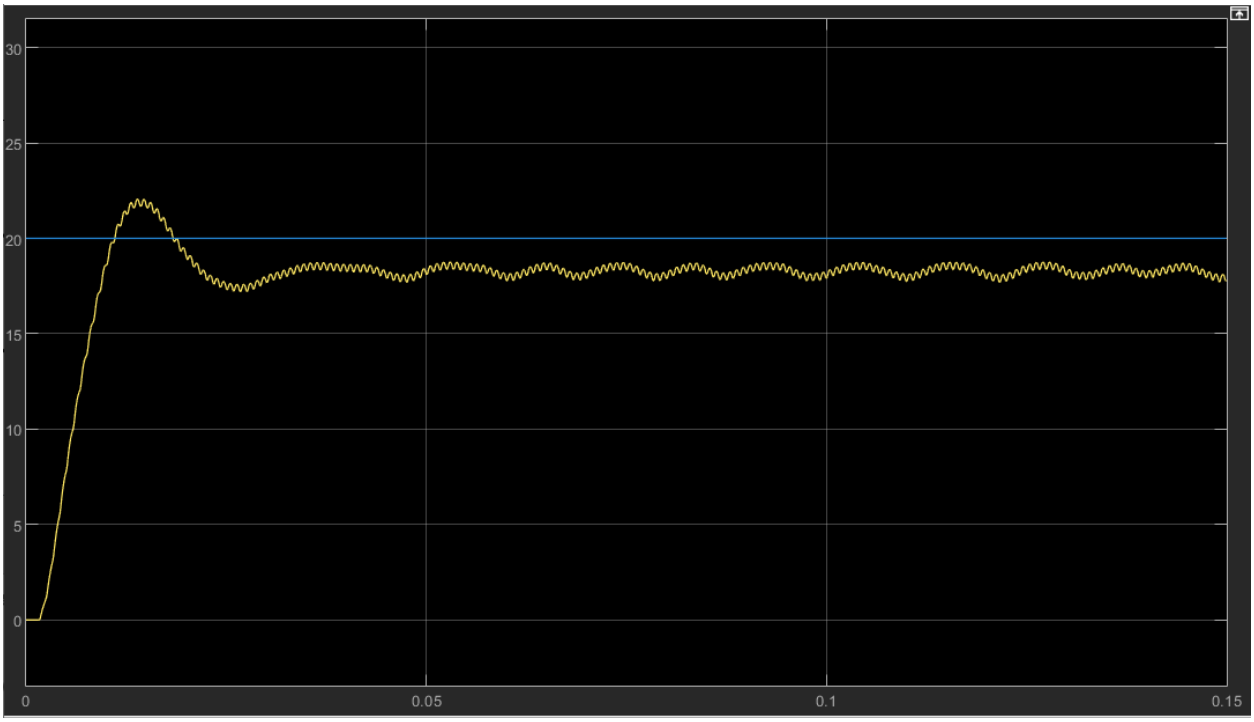


The model is stable (oscilant intretinut) has an overshoot of 54% and settles at 18.3 V with oscillations of ±0.5 V. The ripple is ±0.2V. It reaches a value close to the setpoint of 20V (~1.7 V smaller)

Now I am going to add values of the filteringFactor from 0 to 1 (0, 0.2, 0.5, 0.7 and 1)




30
25
20
15
10
5
0
0
0.05
0.1
0.15


30
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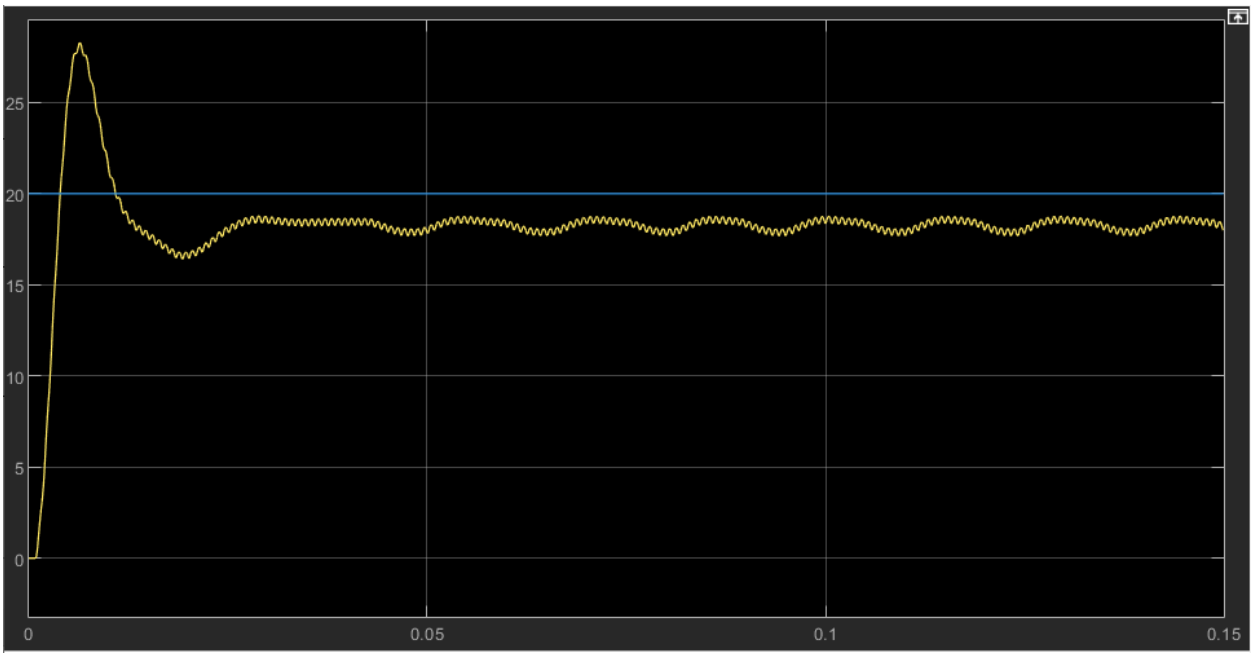

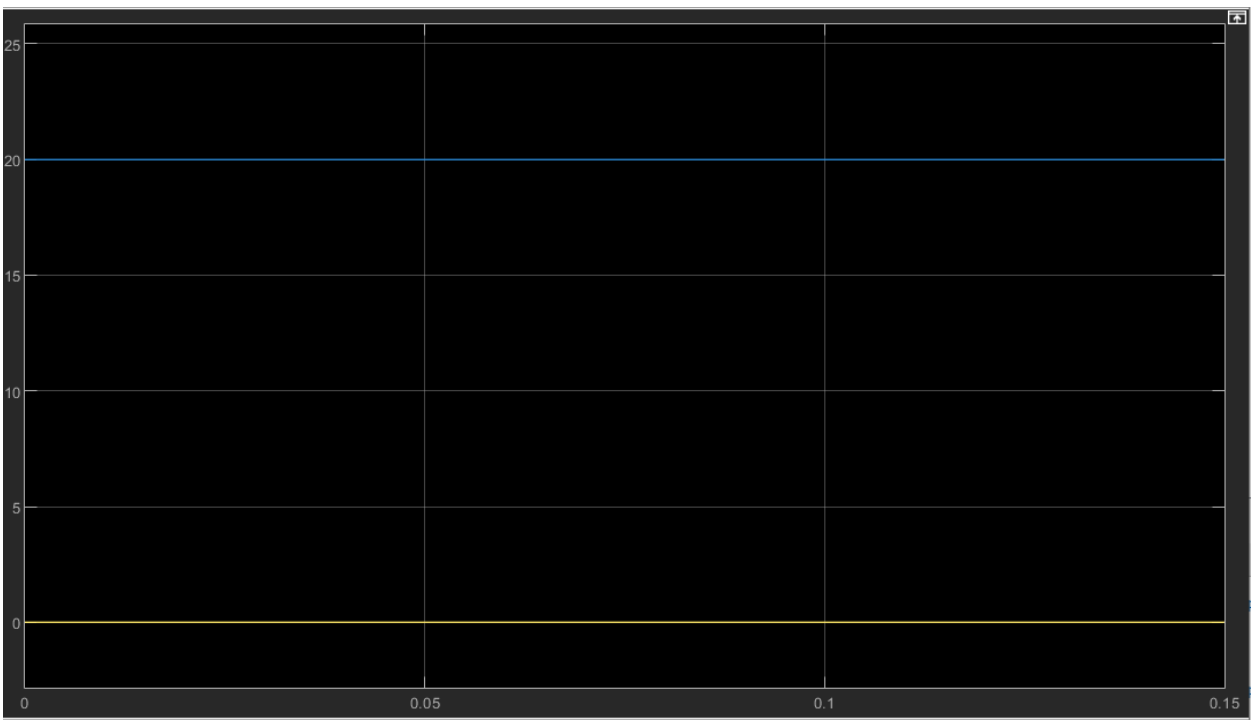

We notice that overshoot may differ for different values and the aspect of the periodic steady state signal may also change a little. In the 1 case we have no iutput at all because none of the input signal is filtered. The period of the steady state signal decreases as the filteringFactor increases.

I will also add the controller function and the plant for extra information

```
Hc =

  160000 s^4 + 2.161e08 s^3 + 1.596e11 s^2 + 6.49e13 s + 1.016e16
  ---------------------------------------------------------------
      1.905e07 s^4 + 3.048e10 s^3 + 1.524e13 s^2 + 2.438e15 s
```

```
Hp =

          1.905e07
  ------------------------
  s^2 + 550.7 s + 3.969e05
```

This is the reccurence equation written in code

```
    //***controler
    static float e1 = 0, e2 = 0, e3 = 0, e4 = 0;
static float y1 = 0, y2 = 0, y3 = 0, y4 = 0;
        // Current input
        float ek = error;   // u[k]

        // Compute output y[k] using recurrence
        float command =
            2.946f  * y1
            - 3.196f  * y2
            + 1.513f  * y3
            - 0.2636f * y4
            + 0.008399f * ek
            - 0.02508f  * e1
            + 0.02935f  * e2
            - 0.01578f  * e3
            + 0.003254f * e4;

        // Shift history (latest moves into y1/e1, oldest is dropped)
        y4 = y3;
        y3 = y2;
        y2 = y1;
        y1 = command;

        e4 = e3;
        e3 = e2;
        e2 = e1;
        e1 = ek;
    //***controler
```